Name: ______________________________ **Date:** _________________ **Class:** _______________

MyDesign Portfolio

Element D Initial Template

How to use this template

Teachers: You may customize this document before providing it to students.

Students: The template below is one way to present your design ideas, but the specific assignment given by your instructor may require a different structure or dictate different details. If you make a copy of this template to edit for your own design ideas, be sure to delete the instructions.

Guidance before filling out the template on the next page:

Because you will iterate on your design multiple times, you will need to create multiple copies of these sections. Think of Element D as a diary that you will date and keep adding to throughout your work until your final iteration. It's critical that iteration shines through in this Element.

You should begin with many design ideas that reflect the individual ideas of your group members and then the group ideas. Number these ideas and include a date, description, drawing, and details for each.

After you have created an initial set of ideas, document your decision making on which design to include, ideally with a design matrix. Create a plan of work, ideally using a Gantt chart that may be housed in this document or in a different day-to-day document. Generate a design blueprint with enough details that someone else not familiar with your project could also build your design.

You will then proceed forward to build and test this design, returning to Element D afterwards when you are ready to create more design ideas. Fill out each of these steps again.

Name: ______________________________ **Date:** _________________ **Class:** _______________

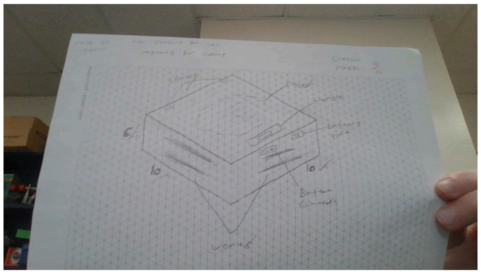
Element D: Design Concept Generation, Analysis, and Selection

Design Idea Generation *(D1)*

This section focuses on the process of generating solutions. The initial rounds should include every idea you come up with but the subsequent rounds of ideation should lead to viable options. Each idea should include not only words, but also a picture. Use as many copies of this sub-template as needed.

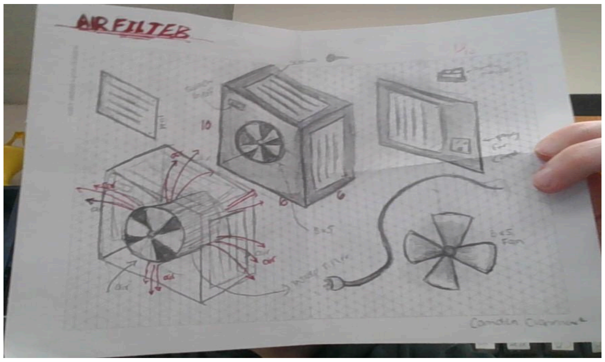
Design Idea #1

Date: 2-22
Description: A plastic box with a fan and filters, it also has a lot of vents for air flow
Drawing that includes distinguishing information such as scale, dimensions, and materials:



Design Idea #2

Date: 2/22/2024
Description: A plastic box with a fan and 4 filters, placed on all sides.
Drawing that includes distinguishing information such as scale, dimensions, and materials:



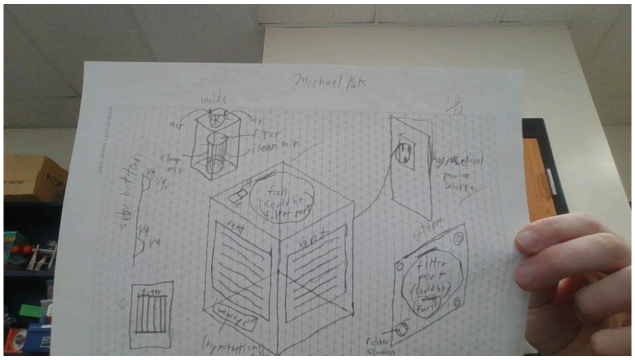
Design Idea #3

Name: ______________________________ **Date:** _________________ **Class:** _______________

Date:2/22/24

Description: a plastic rectangular prism design, with a fan on the top (or bottom), with 4 vents on the sides. The filter will sit inside of the center of the devices. Air will be pushed in through the filter from the top and the purified air will exit from the side vents. 4, ¼ inch tall rubber feet, could be powered by batteries or an outlet cord. Batteries could be rechargeable.

Drawing that includes distinguishing information such as scale, dimensions, and materials:



~~**Design Idea Comparison and Selection**~~ ~~*(D1, D2)*~~

~~When you've finished generating ideas, you must compare your design options according to all your design requirements. Incorporate the priority of those requirements when comparing design options. A decision matrix (perhaps a Pugh Matrix) is encouraged since it allows for systematic, weighted comparisons. If using weights in a decision matrix, be sure to explain and justify the weighting based on your design requirements. You may use the weighted design matrix template below, inserting your criteria and design options.~~

Name: ______________________________ **Date:** _________________ **Class:** _______________

~~\~~

	~~weight~~	~~Option 1~~	~~Option 2~~	~~Option 3~~	~~Option 4~~
~~Criterion A~~					
~~Criterion B~~					
~~Criterion C~~					
~~Criterion D~~					
~~Criterion E~~					
~~score~~					

Design Blueprint *(D4)*

Describe all of your design's details such that someone unfamiliar with your project could build your prototype. Include sketches, dimensions, CAD (if taught in your class), manufacturing methods, material choices, supply lists, and costs.

Design Blueprint #1

Date:
Sketch:
Manufacturing Methods:
    Step 1:
    Step 2:
    Step 3:
Supply List and Costs:

Item	**Cost each**	**# needed**	**Total**
box	$44	1	$44

Name: ______________________________ **Date:** _________________ **Class:** _______________

Filter	$25	1	$25
Fan	$13.99	1	13.99
stickers	$7.99	1	$7.99
wall adapter	$8.00	1	$8.00
USB Extension cord	$7.99	1	$7.99
screws	$7.59	1	$7.59
Total Cost			

Plan of Action *(D3)*

A Gantt chart is encouraged to lay out the primary steps expected on a visual timeline. Larger tasks from the Gantt chart should be broken down further into detailed sub-tasks, which may be housed in a different day-to-day document. Discrete tasks should include enough detail such that others could replicate the efforts, such as procedures, materials, tools, status, etc.

3/6	Start build, make first holes/vents
3/8	Add filter/fan
3/12	Fix mistakes/ solve issues
3/14	Add cord holes and finish the build
3/18	Finish things up/ test
3/20	Final renditions